Oral Disease Classification - Model Comparison (Draft)

Task: classify oral images into 6 classes (Calculus, Caries, Gingivitis, Mouth Ulcer, Tooth Discoloration, hypodontia). Evaluation set: 1,112 images. Classes are imbalanced (Gingivitis 482, hypodontia 257, Calculus 250, Mouth Ulcer 53, Caries 42, Tooth Discoloration 28).

1. Overall results

Model	Accuracy	Macro F1	Weighted F1
Baseline CNN	0.4335	0.10	0.26
EfficientNetB0	0.7950	0.74	0.79
EfficientNetB0 + unfreeze	0.8147	0.77	0.81
ResNet50	0.7581	0.71	0.76
ResNet50 + unfreeze	0.7977	0.76	0.80

2. Per-class F1-score

Class (n)	CNN	Eff	Eff+unf	Res	Res+unf
Calculus (250)	0.00	0.62	0.64	0.63	0.67
Caries (42)	0.00	0.56	0.65	0.47	0.65
Gingivitis (482)	0.60	0.81	0.83	0.75	0.80
Mouth Ulcer (53)	0.00	0.93	0.95	0.92	0.94
Tooth Discoloration (28)	0.00	0.61	0.60	0.55	0.58
hypodontia (257)	0.00	0.95	0.96	0.95	0.95

3. Notes

- The baseline CNN did not learn: it predicts Gingivitis for every image. Its 43% accuracy equals the share of Gingivitis in the data.
- Transfer learning (ImageNet weights) raised accuracy from 43% to 76-81%.
- Unfreezing the last layers helped both models: EfficientNetB0 79.5% to 81.5%, ResNet50 75.8% to 79.8%.
- Mouth Ulcer and hypodontia are easy for all trained models (F1 0.92-0.96).
- Hard classes: Calculus, Caries, Tooth Discoloration (recall of the best model: 0.58, 0.60, 0.54).

4. Best model for deployment

EfficientNetB0 with unfreeze (fine-tuned). It has the best accuracy (81.5%), macro F1 (0.77) and weighted F1 (0.81), and it is much lighter than ResNet50 (about 5M vs 25M parameters), so inference is faster and deployment is easier. ResNet50 + unfreeze is a close second (79.8%).

5. Before deploying

- Check results on a separate test set. If the 1,112 images were also used for checkpointing / early stopping, the scores are a bit optimistic.
- The gap between the top two models (about 1.7 points) is small, so it is not a strong difference by itself.
- Use it as a support tool for a dentist, not as a final diagnosis, because recall on the rare classes is still low (54-60%).
- Improvements: more images for the rare classes, class weights, a confidence threshold for uncertain predictions.